Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
.Net Core and Progressive Web Application

Practical – X

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Class:TYIT	Batch:1
Date of Assignment: 22-09-2026	Date/Time of Submission: 22-09-2026

A) Create a PWA with a Web App Manifest file including:

CODE :-

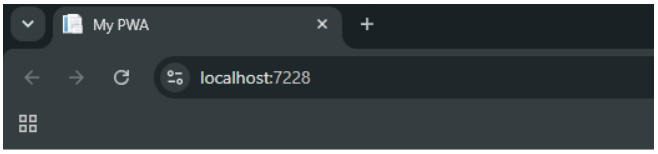
manifest.json :-

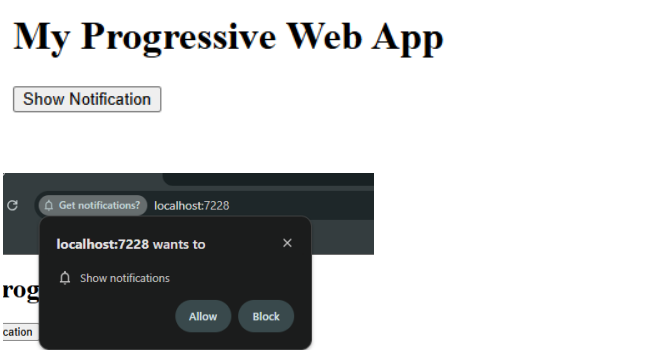
```
{
 "name": "My PWA Application",
 "short_name": "MyPWA",
 "start_url": "/",
 "display": "standalone",
 "background_color": "#ffffff",
 "theme_color": "#0d6efd",
 "icons": [
  {
   "src": "icons/icon1.png",
   "sizes": "192x192",
   "type": "image/png"
  },
  {
   "src": "icons/icon2.png",
   "sizes": "512x512",
   "type": "image/png"
  }
 ]
}
```

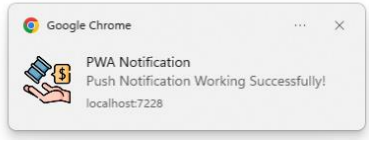
index.html :-

```
<!DOCTYPE html>
<html>
<head>
   <title>My PWA</title>
   <link rel="manifest" href="manifest.json">
   <meta name="theme-color" content="#0d6efd">
</head>
<body>
   <h1>My Progressive Web App</h1>
   <button onclick="showNotification()">
      Show Notification
   </button>
   <script src="app.js"></script>
</body>
```

OUTPUT :-







```
</html>
```

app.js :-

```
if ('serviceWorker' in navigator) {
   navigator.serviceWorker.register('service-worker.js')
      .then(() => console.log('Service Worker Registered'));
}

function showNotification() {
   Notification.requestPermission().then(permission => {
      if (permission === 'granted') {
         navigator.serviceWorker.ready.then(registration => {
            registration.showNotification('PWA Notification', {
               body: 'Push Notification Working Successfully!',
               icon: 'icons/icon1.png'
            });
         });
      }
   });
}
```

service-worker.js :-

```
if ('serviceWorker' in navigator) {
   navigator.serviceWorker.register('service-worker.js')
      .then(() => console.log('Service Worker Registered'));
}
function showNotification() {
   Notification.requestPermission().then(permission => {
      if (permission === 'granted') {
         navigator.serviceWorker.ready.then(registration => {
            registration.showNotification('PWA Notification', {
               body: 'Push Notification Working Successfully!',
               icon: 'icons/icon1.png'
            });
         });
      }
   });
}
```

Program.cs :-

```
var builder = WebApplication.CreateBuilder(args);
var app = builder.Build();
app.UseDefaultFiles();
app.UseStaticFiles();
app.Run();
```

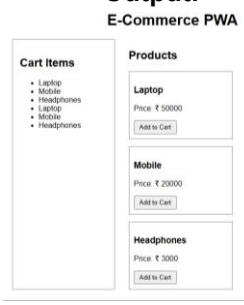
B) Develop an E-Commerce PWA with Products Listing and Offline Cart Personality

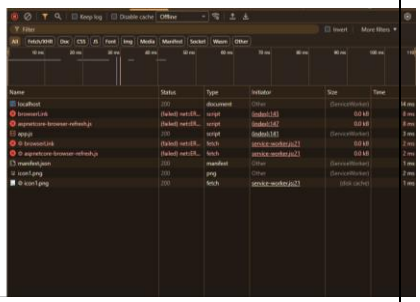
CODE :-

Manifest.json:-

```
{
  "name": "E-Commerce PWA",
  "short_name": "ShopPWA",
  "id": "/",
  "start_url": "/",
```

Output:-

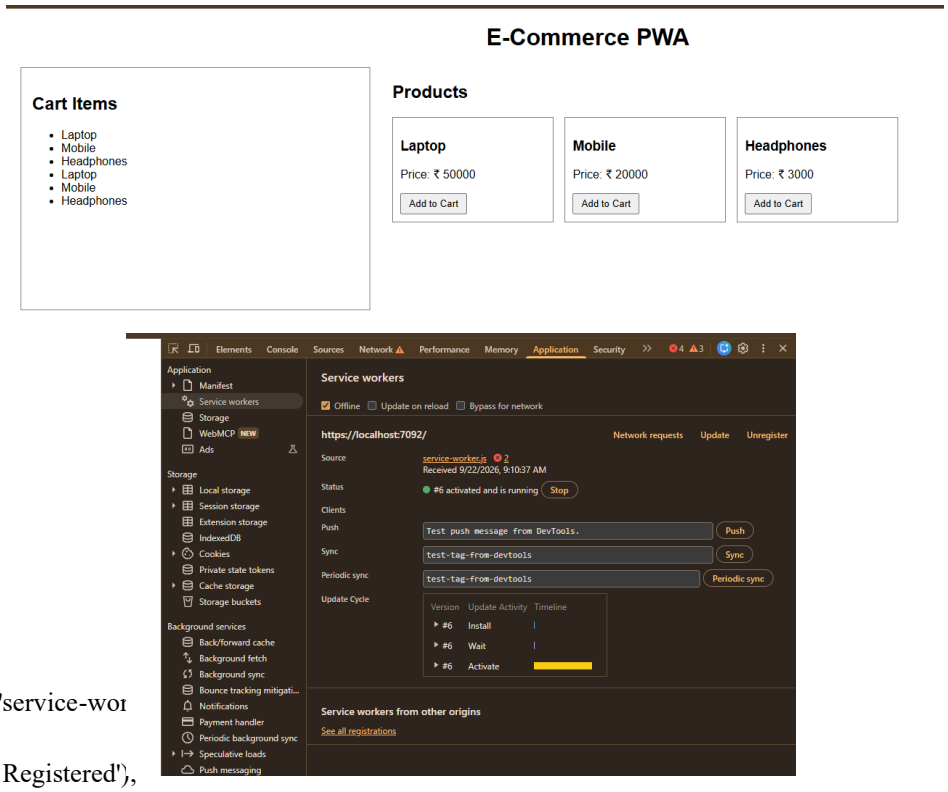




```
  "scope": "/",
  "display": "standalone",
  "background_color": "#ffffff",
  "theme_color": "#0d6efd",
  "icons": [
    {
      "src": "/icons/icon1.png",
      "sizes": "128x128",
      "type": "image/png"
    },
    {
      "src": "/icons/icon2.png",
      "sizes": "512x512",
      "type": "image/png"
    }
  ]
}
```

app.js:-

```
// Register Service Worker
if ('serviceWorker' in navigator) {
    navigator.serviceWorker.register('service-wor
        .then(() => {
            console.log('Service Worker Registered'),
        });
}
// Get cart from Local Storage
let cart = JSON.parse(localStorage.getItem('cart')) || [];
// Display cart when page loads
displayCart();
// Add product to cart
function addToCart(product) {
    cart.push(product);
    localStorage.setItem('cart', JSON.stringify(cart));
    displayCart();
}
// Display cart items
function displayCart() {
    let cartList = document.getElementById('cartList');
    cartList.innerHTML = '';
    cart.forEach(item => {
        let li = document.createElement('li');
        li.textContent = item;
        cartList.appendChild(li);
        if ("serviceWorker" in navigator) {
            navigator.serviceWorker.register("/service-worker.js")
                .then(function () {
                    console.log("Service Worker Registered");
                })
                .catch(function (error) {
```



```
                console.log("Service Worker Error:", error);
            });
        }
    });
}
```

service-worker.js:

```
const CACHE_NAME = "ecommerce-pwa-v1";
const FILES_TO_CACHE = [
    "/",
    "/index.html",
    "/app.js",
    "/manifest.json"
];
self.addEventListener("install", event => {
    event.waitUntil(
        caches.open(CACHE_NAME)
            .then(cache => cache.addAll(FILES_TO_CACHE))
    );
});
self.addEventListener("fetch", event => {
    event.respondWith(
        caches.match(event.request)
            .then(response => {
                return response || fetch(event.request);
            })
    );
});
```

Appsetting.json:-

```
{
  "Logging": {
    "LogLevel": {
      "Default": "Information",
      "Microsoft.AspNetCore": "Warning"
    }
  },
  "AllowedHosts": "*"
}
```

Index.html:-

```
<!DOCTYPE html>
<html>
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>E-Commerce PWA</title>
    <link rel="manifest" href="manifest.json">
    <meta name="theme-color" content="#0d6efd">
    <style>
        body {
            font-family: Arial;
```

```
            margin: 20px;
        }
        h1 {
            text-align: center;
        }
        .container {
            display: flex;
            gap: 30px;
        }
        .cart-section {
            width: 30%;
            border: 1px solid gray;
            padding: 15px;
            min-height: 300px;
        }
        .product-section {
            width: 70%;
        }
        .products {
            display: flex;
            flex-wrap: wrap;
            gap: 15px;
        }
        .product {
            border: 1px solid gray;
            padding: 10px;
            width: 200px;
        }
        button {
            padding: 5px 10px;
            cursor: pointer;
        }
    </style>
    <link rel="manifest" href="/manifest.json">
    <meta name="theme-color" content="#0d6efd">
</head>
<body>
    <h1>E-Commerce PWA</h1>
    <div class="container">
        <!-- Cart Section -->
        <div class="cart-section">
            <h2>Cart Items</h2>
            <ul id="cartList"></ul>
        </div>
        <!-- Product Section -->
        <div class="product-section">
            <h2>Products</h2>
            <div class="products">
                <!-- Laptop -->
```

```
            <div class="product">
                <h3>Laptop</h3>
                <p>Price: ₹ 50000</p>
                <button onclick="addToCart('Laptop')">
                    Add to Cart
                </button>
            </div>
            <!-- Mobile -->
            <div class="product">
                <h3>Mobile</h3>
            <p>Price: ₹ 20000</p>
                <button onclick="addToCart('Mobile')">
                    Add to Cart
                </button>
            </div>
            <!-- Headphones -->
            <div class="product">
                <h3>Headphones</h3>
                <p>Price: ₹ 3000</p>
                <button onclick="addToCart('Headphones')">
                    Add to Cart
                </button>
            </div>
        </div>
    </div>
  </div>
  <script src="app.js"></script>
</body>
</html>
```

program.cs:-

```
var builder = WebApplication.CreateBuilder(args);
var app = builder.Build();
app.UseDefaultFiles();
app.UseStaticFiles();
app.Run();
```